

Iron-Catalyzed Aerobic Oxidation of Alcohols: Lower Cost and Improved Selectivity

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Supporting Information

ABSTRACT: An aerobic oxidation reaction of alcohols toward aldehydes or ketones using catalytic amounts of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 4-OH-TEMPO, and NaCl has been developed. Compared with the former catalytic system with TEMPO developed in this group, the new protocol using 4-OH-TEMPO, which is much cheaper on an industrial scale, accomplished the transformation with a higher selectivity, especially for aliphatic alcohols toward aldehydes. α,β -Unsaturated alkynals or alkynones can be efficiently synthesized from propargyl alcohols, which has been much less studied in the literature.

KEYWORDS: aerobic oxidation, iron catalysis, 4-OH-TEMPO, aldehydes, ketones

INTRODUCTION

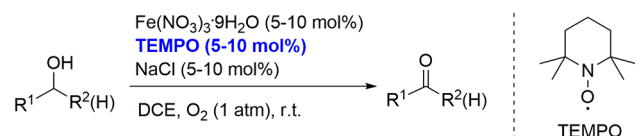
Aldehydes and ketones are important chemicals that are widely used in daily life, industry, and academic research.¹ Oxidation of alcohols is a direct and significant way to produce aldehydes and ketones.² Traditionally, stoichiometric amounts of oxidants containing heavy metals (Mn,³ Cr,⁴ etc.) or expensive hypervalent iodines⁵ such as Dess–Martin reagent or IBX reagent have been used in industry and the laboratory. Besides the high cost, these methods usually produce large amounts of toxic byproducts during the production process, causing severe environmental burdens, which makes it necessary to develop green oxidation methods using molecular oxygen, a clean, cheap, and sustainable substance, as the oxidant.⁶ Early reports focused on catalysis by noble metals such as Pd,⁷ Ru,⁸ Au,⁹ Ag,¹⁰ etc. The high cost, limited substrate scope, and harsh conditions are main problems to be solved. In recent years, aerobic oxidation of alcohols using earth-abundant catalysts such as Fe,^{11–13} Cu,¹⁴ and Co¹⁵ has received extensive attention. In 2011, our group developed a highly efficient $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO/NaCl-catalyzed aerobic oxidation reaction of alcohols to aldehydes or ketones under mild conditions with broad substrate scope (Scheme 1).¹² $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and NaCl are both very cheap chemicals, while TEMPO, though much cheaper than many other *N*-oxyls such as ABNO and AZADO, is still relatively expensive for industrial-scale production. 4-OH-TEMPO is a much cheaper *N*-oxyl even than TEMPO.¹⁶ In this work, we used 4-OH-TEMPO instead of TEMPO as the cocatalyst based on our previous catalytic system for a more cost-effective reaction (Scheme 1).

RESULTS AND DISCUSSION

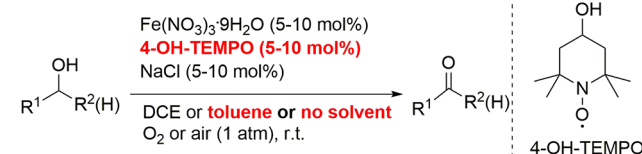
It should be noticed that the chloride anion is used as the ligand to increase the activity of the iron catalyst^{12a} and that the cation of the inorganic chloride plays an important role in steps of

Scheme 1. Iron-Catalyzed Aerobic Oxidations of Alcohols to Aldehydes or Ketones

Previous work of this group:



This work:



ligand exchange.^{17a} We started our research using 1-dodecanol (**1a**) as the initial substrate for the aerobic oxidation reaction. With 5 mol % loadings of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 4-OH-TEMPO, and NaCl as the catalysts, the reaction in DCE at 25 °C for 20.5 h using molecular oxygen as the oxidant gave aldehyde **2a** in 63% NMR yield as well as 27% NMR recovery of alcohol **1a** (Table 1, entry 1). When the reaction time was extended to 28.5 h, the NMR yield of aldehyde **2a** was improved to 71%, with 14% NMR recovery of alcohol **1a** (Table 1, entry 2). When the loadings of 4-OH-TEMPO and NaCl were increased to 10 mol %, an 85% NMR yield of aldehyde **2a** was formed along with an 11% NMR yield of acid **3a** after 20 h (Table 1, entry 3). We then shortened the reaction time to 19 h, and the NMR yield of aldehyde **2a** was 88% with only 3% NMR yield of acid **3a** and 2.5% NMR recovery of alcohol **1a** (Table 1, entry 4). A 57%

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Table 1. Optimization of the Aerobic Oxidation Reaction of **1a**^a

$ \begin{array}{c} \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (x mol\%)} \\ \text{4-OH-TEMPO (y mol\%)} \\ \text{NaCl (z mol\%)} \\ \hline n\text{-C}_{11}\text{H}_{23}\text{CH}_2\text{OH} \xrightarrow{\text{DCE, O}_2 \text{ bag, 25 }^\circ\text{C, t (h)}} n\text{-C}_{11}\text{H}_{23}\text{CHO} + n\text{-C}_{11}\text{H}_{23}\text{COOH} \\ \text{1a} \qquad \qquad \qquad \text{2a} \qquad \qquad \qquad \text{3a} \end{array} $							
Entry	x	y	z	t (h)	Yield (%) ^b		Recovery of 1a (%) ^b
					2a	3a	
1	5	5	5	20.5	63	trace	27
2	5	5	5	28.5	71	1	14
3	5	10	10	20	85	11	—
4	5	10	10	19	88	3	2.5
5	5	10	0	24	57	—	34
6	10	10	10	9	85	10	1.5
7 ^c	10	10	10	12	78	21	—
8 ^{c,d}	10	10	10	12	—	100	—

^aThe reactions were carried out on a 1.0 mmol scale of **1a** in 4.0 mL of DCE with a bag of O₂. ^bNMR yields. ^cKCl was used instead of NaCl.

^dTEMPO was used instead of 4-OH-TEMPO.

NMR yield of aldehyde **2a** with 34% NMR recovery of substrate **1a** was observed in the reaction for 24 h in the absence of NaCl, indicating a significant effect of NaCl for accelerating the reaction (Table 1, entry 5). We attempted to use 10 mol % loadings of Fe(NO₃)₃·9H₂O, 4-OH-TEMPO, and NaCl to shorten the reaction time. However, a 10% NMR yield of acid **3a** was already observed in 9 h along with an 85% NMR yield of aldehyde **2a** and 1.5% NMR recovery of substrate **1a** (Table 1, entry 6). We even tried to accomplish the aerobic oxidation of alcohol **1a** toward carboxylic acid **3a** with 10 mol % loadings of Fe(NO₃)₃·9H₂O, 4-OH-TEMPO, and KCl,¹⁷ but aldehyde **2a** remained as the main product in 78% NMR yield, and a 21% NMR yield of acid **3a** was formed (Table 1, entry 7). In our previous report,¹⁷ the transformation of alcohol **1a** to acid **3a** was completed in 12 h (Table 1, entry 8), indicating that the reaction using 4-OH-TEMPO has a higher selectivity for the formation of aldehydes.

Then we conducted a screening of other solvents. The reactions in CH₃CN, THF, EtOAc, and dioxane for 24 h gave only 18–31% NMR yields of aldehyde **2a** (Table 2, entries 1–4). The reaction proceeded smoothly in other chlorinated solvents such as DCM and CHCl₃, with ~30% NMR recovery of alcohol **1a** after 24 h (Table 2, entries 5 and 6). In toluene, the reaction gave a 77% NMR yield of aldehyde **2a** along with 14%

NMR recovery of alcohol **1a** (Table 2, entry 7). Thus, DCE is the optimal solvent.

With the standard conditions in hand, we investigated the oxidation reaction of different alcohols using molecular oxygen as the oxidant. Aliphatic alcohols could be oxidized to aldehydes with high selectivity (Table 3, entries 1 and 2). Aliphatic alcohol **1c** with an ester group was transformed into aldehyde **2c** smoothly (Table 3, entry 3). Allylic alcohol **E-1d** and propargyl alcohol **1e** could also be oxidized to the corresponding aldehydes with high selectivity (Table 3, entries 4 and 5). For benzylic alcohols, which are usually more reactive in such reactions, only 5 mol % loadings of Fe(NO₃)₃·9H₂O, 4-OH-TEMPO, and NaCl were required with a much shorter reaction time. For example, 4-nitrobenzyl alcohol (**1f**) was converted into 4-nitrobenzaldehyde (**2f**) in 91% yield in 5 h. According to the guidelines for residual solvents in the pharmaceutical industry (ICH Q3C), 1,2-dichloroethane is listed as a hazardous solvent to be avoided (concentration limit: 5 ppm), whereas toluene is much safer (concentration limit: 890 ppm). Thus, we also carried out the oxidation of **E-1d** using toluene as the solvent, which afforded **E-2d** in 92% yield (Table 3, entry 7).

The reaction could also be conducted under air conditions. Benzylic alcohols **1g–k** with different substituents could be oxidized into aldehydes in high yields with high selectivity in 5–9 h (Table 4, entries 1–4). A variety of synthetically useful functional groups such as iodo, chloro, methoxy, ester, and nitro groups were well-tolerated in the reaction. Interestingly, for biomass-based substrate 5-hydroxymethylfurfural (**1k**, HMF),¹⁸ the reaction produced 2,5-diformylfuran (**2k**, DFF) exclusively in 90% yield. The reaction could also be applied to the selective oxidation of terminal alkynol **1l** and allenol **1m** to the corresponding aldehydes when the catalyst loadings were increased.

The oxidation of propargyl alcohols to aldehydes and ketones using molecular oxygen as the oxidant has been less studied in the literature. We investigated different primary and secondary propargyl alcohols. For primary propargyl alcohols, apart from alkyl-substituted propargyl alcohol **1e** (Table 3, entry 5), aryl-substituted propargyl alcohols could also be oxidized to aldehydes with a high selectivity (Table 5, entries 1 and 2). For substrate **1o** with an electron-donating OMe group on the phenyl ring, aldehyde **2o** was obtained in 77% yield using 10 mol % loadings of the catalysts (Table 5, entry 2). Aryl- and

Table 2. Screening of Solvents for the Aerobic Oxidation Reaction of **1a**^a

$ \begin{array}{c} \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (5 mol\%)} \\ \text{4-OH-TEMPO (10 mol\%)} \\ \text{NaCl (10 mol\%)} \\ \hline n\text{-C}_{11}\text{H}_{23}\text{CH}_2\text{OH} \xrightarrow{\text{solvent, O}_2 \text{ bag, 25 }^\circ\text{C, 24 h}} n\text{-C}_{11}\text{H}_{23}\text{CHO} \\ \text{1a} \qquad \qquad \qquad \text{2a} \end{array} $			
Entry	Solvent	Yield of 2a (%) ^b	Recovery of 1a (%) ^b
1	CH ₃ CN	31	61
2	THF	18	76
3	EtOAc	22	70
4	dioxane	24	70
5	DCM	65	28
6	CHCl ₃	65	31
7	toluene	77	14

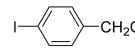
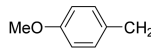
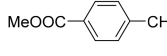
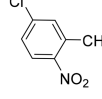
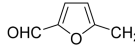
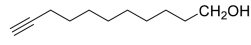
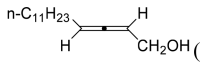
^aThe reactions were carried out on a 1.0 mmol scale of **1a** in 4.0 mL of solvent with a bag of O₂. ^bNMR yields.

Table 3. Scope of the Aerobic Oxidation Reaction of Alcohols under O₂ Conditions^a

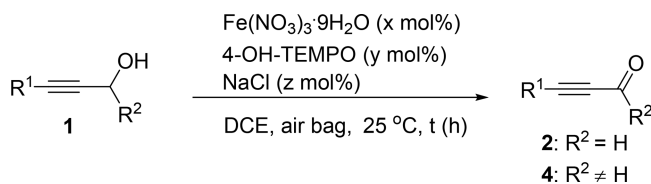
		Fe(NO ₃) ₃ ·9H ₂ O (x mol%) 4-OH-TEMPO (y mol%) NaCl (z mol%)				RCHO 2	+	RCOOH 3	
RCH ₂ OH 1		DCE, O ₂ bag, 25 °C, t (h)							
Entry	Substrate	x	y	z	t (h)	Yield		Recovery of 1 (%) ^c	
						2 (%) ^b	3 (%) ^c		
1	<i>n</i> -C ₁₁ H ₂₃ CH ₂ OH (1a)	5	10	10	19	89 (2a)	3	2.5	
2	<i>n</i> -C ₁₅ H ₃₁ CH ₂ OH (1b)	5	10	10	16	87 (2b)	trace	3	
3	MeOOC(CH ₂) ₆ OH (1c)	5	10	10	16	63 (2c)	3.5	-	
4	 (E-1d)	5	10	10	19	87 (E-2d)	-	-	
5	 (1e)	5	5	5	13	92 (2e)	2	-	
6	 (1f)	5	5	5	5	91 (2f)	1.5	-	
7 ^d	E-1d	5	10	10	24	92 (E-2d)	-	-	

^aThe reactions were carried out on a 1.0 mmol scale of **1** in 4.0 mL of DCE with a bag of O₂. ^bIsolated yields. ^cNMR yields. ^dToluene was used instead of DCE.

Table 4. Scope of the Aerobic Oxidation Reaction of Alcohols under Air Conditions^a

		Fe(NO ₃) ₃ ·9H ₂ O (x mol%) 4-OH-TEMPO (y mol%) NaCl (z mol%)				RCHO 2	
Entry	Substrate	x	y	z	t (h)	Yield of 2 (%) ^b	
1	 (1g)	5	5	5	5	99 (2g)	
2	 (1h)	5	5	5	9	98 (2h)	
3	 (1i)	5	5	5	6	95 (2i)	
4	 (1j)	5	5	5	6	99 (2j)	
5	 (1k)	5	5	5	9	90 (2k)	
6 ^c	 (1l)	10	10	10	11.5	79 (2l)	
7	 (1m)	5	10	10	12	77 (2m)	

^aThe reactions were carried out on a 1.0 mmol scale of **1** in 4.0 mL of DCE with a bag of air. ^bIsolated yields. ^c6% NMR yield of 10-undecynoic acid and 2% NMR recovery of the substrate.

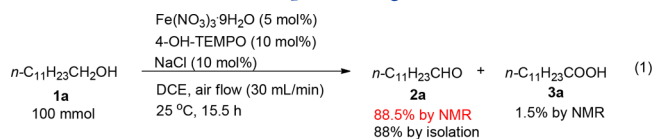
Table 5. Scope of Propargyl Alcohols under Air Conditions^a

Entry	R ¹ , R ² (1)	x	y	z	t (h)	Yield of 2 or 4 (%) ^b
1	Ph, H (1n)	5	5	5	7	81 (2n)
2	4-MeOC ₆ H ₄ , H (1o)	10	10	10	9.5	77 (2o)
3	H, Ph (1p)	5	5	5	4	89 (4p)
4	<i>n</i> -C ₄ H ₉ , <i>n</i> -C ₇ H ₁₅ (1q)	5	5	5	7	92 (4q)
5	<i>n</i> -C ₄ H ₉ , 4-NCC ₆ H ₄ (1r)	5	5	5	7	91 (4r)
6 ^c	H, Ph (1p)	5	5	5	8	85 (4p)

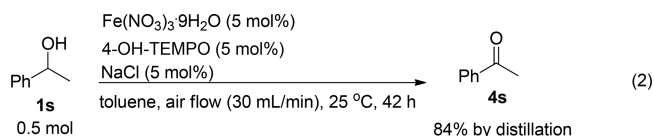
^aThe reactions were carried out on a 1.0 mmol scale of **1** in 4.0 mL of DCE with a bag of air. ^bIsolated yields. ^cToluene was used instead of DCE.

alkyl-substituted secondary propargyl alcohols **1p–r** could be oxidized to ketones in high yields (Table 5, entries 3–5). The cyano group was well-tolerated in the reaction (Table 5, entry 5). To our delight, **1p** could be oxidized to **4p** in toluene using air as the terminal oxidant in a comparable yield in 8 h (Table 5, entry 6).

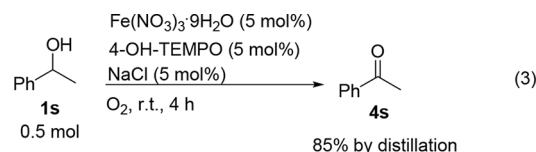
In order to show the practicality in an academic synthetic laboratory, a 100 mmol scale reaction of **1a** applying a slow flow of air was demonstrated (eq 1 and Figure 1).



In addition, we conducted a 0.5 mol scale reaction of **1s** in toluene with a slow flow of air, and the corresponding product **4s** was conveniently isolated by simple distillation in 84% yield (eq 2).



Further study showed that the liquid **1s** could also be oxidized to **4s** with 85% isolated yield in 4 h under the atmosphere of oxygen without any solvent (eq 3).



CONCLUSIONS

An iron-catalyzed aerobic oxidation reaction of alcohols to aldehydes and ketones using Fe(NO₃)₃·9H₂O, 4-OH-TEMPO, and NaCl as the catalyst system showed improved selectivity for the formation of aldehydes compared with the catalyst system using the more expensive TEMPO. The reaction may be conducted under either O₂ or air conditions at room temperature with broad substrate scope, and reactions on up to 0.5 mol scale were demonstrated. Further studies in this area are being actively pursued in this laboratory.

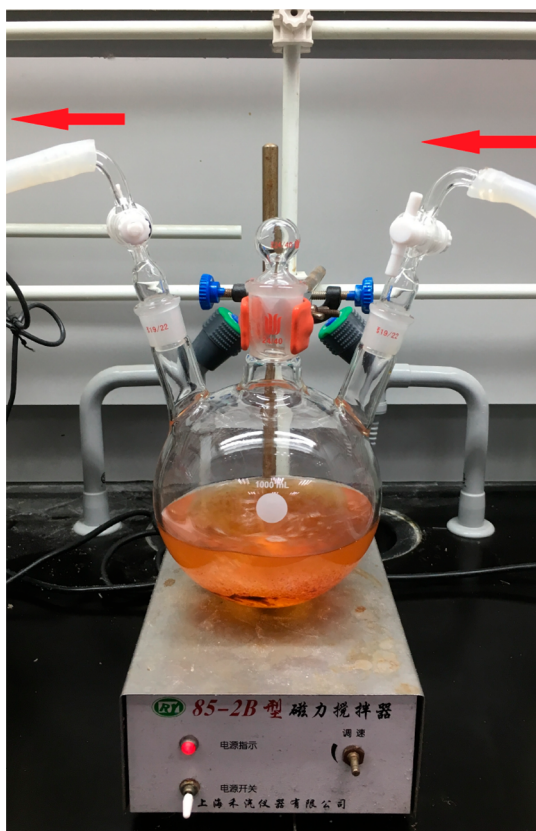
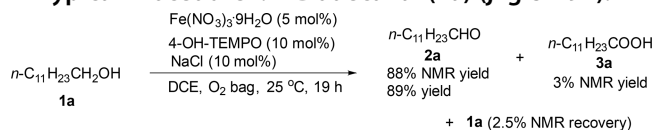


Figure 1. The apparatus and reaction for the aerobic reaction of **1a** on a 100 mmol scale with a slow flow of air (30 mL/min).

EXPERIMENTAL SECTION

General Information. $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was purchased from Energy Chemical or J&K. 4-OH-TEMPO was purchased from Adamas. NaCl was purchased from Sinopharm. DCE was used directly without further treatment. Other reagents were used as received without further treatment. Substrates **1c**,²² **1e**,²⁶ **1l**,³⁶ **1m**,³⁹ **1o**,⁴⁰ **1q**,⁴² and **1r**⁴⁴ were synthesized according to the literature methods. Petroleum ether (60–90 °C) was used for chromatography. Gas bags for reactions with O_2 or air were obtained from Wattcas. Melting points were recorded using a Stuart SMP30 melting point apparatus. IR spectra were obtained using a Bruker Tensor 27 spectrometer. ^1H and ^{13}C NMR spectra were obtained using an Agilent (400 MHz), Varian Mercury (400 MHz), or Bruker (400 MHz) spectrometer. Mass spectra were recorded using an Agilent 5973N instrument. **Caution:** Oxygen is used in combination with organic solvents; all ignition sources should be removed, including sources of sparks, static, or flames, since oxygen increases the intensity of any fire. Inhalation of pure oxygen should be avoided as well.

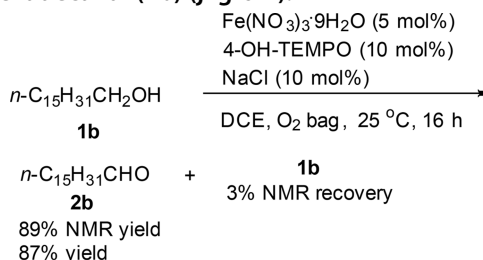
Typical Procedure I: *n*-Dodecanal (**2a**) (jxg-5-192).



To a Schlenk tube were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.3 mg, 0.05 mmol), 4-OH-TEMPO (17.2 mg, 0.1 mmol), NaCl (5.9 mg, 0.1 mmol), **1a** (188.0 mg, 99% purity, 1.0 mmol), and DCE (4.0 mL) sequentially under an atmosphere of oxygen from a gas bag (2 L (commercial size); could be expanded to 5 L). The Schlenk tube was then stirred at 25 °C until completion of the reaction as

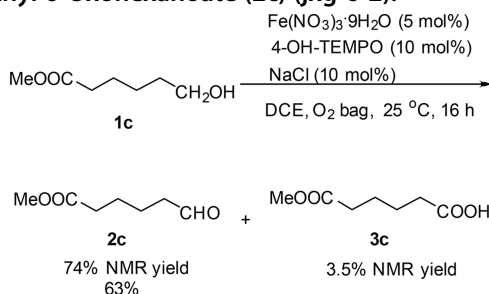
monitored by TLC (petroleum ether/ethyl acetate = 5/1) (19 h). The crude reaction mixture was filtered through a short column of silica gel (height, 2 cm; Φ , 3 cm), eluting with diethyl ether (3 $\times$ 25 mL). After evaporation, the residue was purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 100/1) to afford **2a**¹⁹ (164.6 mg, 89%) as an oil (88% **2a**, 3% **3a**,²⁰ and 2.5% **1a** were observed by NMR analysis of the crude product using CH_2Br_2 as an internal standard). ^1H NMR (400 MHz, CDCl_3) δ 9.77 (t, J = 1.8 Hz, 1H, CHO), 2.42 (td, J_1 = 7.3 Hz, J_2 = 2.1 Hz, 2H, CH_2), 1.63 (quint, J = 7.2 Hz, 2H, CH_2), 1.38–1.20 (m, 16H, 8 $\times$ CH_2), 0.88 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, CDCl_3) δ 202.9, 43.9, 31.9, 29.6, 29.5, 29.4, 29.33, 29.29, 29.1, 22.6, 22.0, 14.1; IR (neat, cm^{-1}) 2922, 2853, 2713, 1726, 1464, 1411, 1389; MS (EI, 70 eV) m/z (%) 184 (M^+ , 0.19), 57 (100).

n-Hexadecanal (**2b**) (jxg-6-4).



Following Typical Procedure I, the reaction of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.4 mg, 0.05 mmol), 4-OH-TEMPO (17.3 mg, 0.1 mmol), NaCl (5.8 mg, 0.1 mmol), and **1b** (247.2 mg, 98% purity, 1.0 mmol) in DCE (4.0 mL) for 16 h afforded **2b**²¹ (209.9 mg, 87%) as a solid (eluent: petroleum ether/ethyl acetate = 100/1 (500 mL)) (89% **2b** and 3% **1b** were observed by NMR analysis of the crude product using CH_2Br_2 as an internal standard). Mp 35.1–36.4 °C (ethyl acetate/petroleum ether; lit.²¹ 35–36 °C); ^1H NMR (400 MHz, CDCl_3) δ 9.76 (t, J = 1.8 Hz, 1H, CHO), 2.42 (td, J_1 = 7.2 Hz, J_2 = 2.0 Hz, 2H, CH_2), 1.63 (quint, J = 7.2 Hz, 2H, CH_2), 1.37–1.19 (m, 24H, 12 $\times$ CH_2), 0.88 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, CDCl_3) δ 202.9, 43.9, 31.9, 29.67, 29.65, 29.64, 29.63, 29.61, 29.55, 29.4, 29.3, 29.1, 22.7, 22.0, 14.1; IR (neat, cm^{-1}) 2913, 2848, 2750, 1704, 1470, 1410, 1392, 1372, 1067; MS (EI, 70 eV) m/z (%) 240 (1.32), 82 (100).

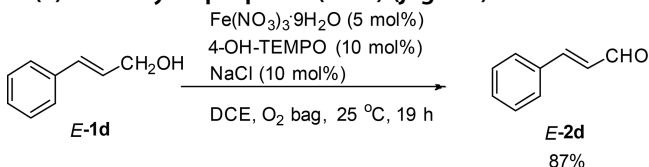
Methyl 6-Oxohexanoate (**2c**) (jxg-6-2).



Following Typical Procedure I, the reaction of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.3 mg, 0.05 mmol), 4-OH-TEMPO (17.4 mg, 0.1 mmol), NaCl (5.8 mg, 0.1 mmol), and **1c**²² (146.0 mg, 1.0 mmol) in DCE (4.0 mL) for 16 h afforded **2c**²³ (91.1 mg, 63%) as an oil (eluent: petroleum ether/ethyl acetate = 15/1 (800 mL)) (74% **2c** and 3.5% **3c**²⁴ were observed by NMR analysis of the crude product using CH_2Br_2 as an internal standard). ^1H NMR (400 MHz, CDCl_3) δ 9.78 (t, J = 1.6 Hz, 1H, CHO), 3.68 (s, 3H, CH_3), 2.52–2.44 (m, 2H, CH_2), 2.39–2.32 (m, 2H, CH_2), 1.72–1.62 (m, 4H, 2 $\times$ CH_2); ^{13}C NMR (100 MHz, CDCl_3) δ

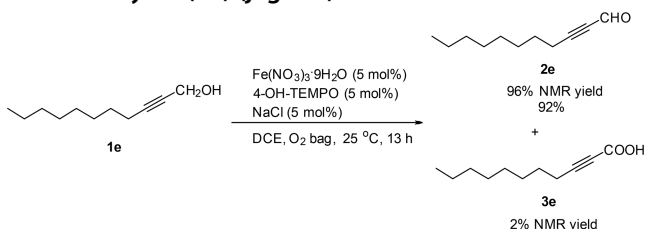
202.0, 173.6, 51.5, 43.4, 35.6, 24.2, 21.4; IR (neat, cm^{-1}) 2952, 2725, 1721, 1437, 1365, 1197, 1155, 1093, 1008; MS (EI, 70 eV) m/z (%) 144 (M^+ , 0.57), 114 (100).

(E)-3-Phenyl-2-propenal (E-2d) (jxg-6-1).



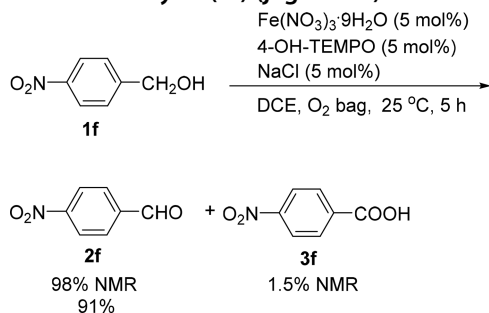
Following Typical Procedure I, the reaction of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.4 mg, 0.05 mmol), 4-OH-TEMPO (17.3 mg, 0.1 mmol), NaCl (5.9 mg, 0.1 mmol), and **1d** (135.0 mg, 99% purity, 1.0 mmol) in DCE (4.0 mL) for 19 h afforded **E-2d**²⁵ (114.8 mg, 87%) as an oil (eluent: petroleum ether/ethyl ether = 20/1 (600 mL)). ^1H NMR (400 MHz, CDCl_3) δ 9.70 (d, J = 7.6 Hz, 1H, CHO), 7.60–7.53 (m, 2H, Ar–H), 7.51–7.40 (m, 4H, 3 $\times$ Ar–H and =CH), 6.72 (dd, J_1 = 16.2 Hz, J_2 = 7.8 Hz, 1H, =CH); ^{13}C NMR (100 MHz, CDCl_3) δ 193.7, 152.8, 133.9, 131.2, 129.0, 128.5, 128.4; IR (neat, cm^{-1}) 2812, 2741, 1668, 1623, 1574, 1493, 1449, 1393, 1328, 1294, 1249, 1119, 1072, 1006; MS (EI, 70 eV) m/z (%) 132 (M^+ , 59.43), 131 (100).

2-Undecynal (2e) (jxg-6-3).



Following Typical Procedure I, the reaction of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.2 mg, 0.05 mmol), 4-OH-TEMPO (8.7 mg, 0.05 mmol), NaCl (3.0 mg, 0.05 mmol), and **1e**²⁶ (167.6 mg, 1.0 mmol) in DCE (4.0 mL) for 13 h afforded **2e**²⁷ (152.9 mg, 92%) as an oil (eluent: petroleum ether/ethyl acetate = 100/1 (400 mL)) (96% **2e** and 2% **3e**²⁸ were observed by NMR analysis of the crude product using CH_2Br_2 as an internal standard). ^1H NMR (400 MHz, CDCl_3) δ 9.18 (s, 1H, CHO), 2.41 (t, J = 7.0 Hz, 2H, CH_2), 1.60 (quint, J = 7.4 Hz, 2H, CH_2), 1.46–1.35 (m, 2H, CH_2), 1.35–1.20 (m, 8H, 4 $\times$ CH_2), 0.89 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, CDCl_3) δ 177.2, 99.3, 81.6, 31.7, 29.0, 28.9, 28.8, 27.5, 22.6, 19.1, 14.0; IR (neat, cm^{-1}) 2925, 2855, 2280, 2199, 1669, 1464, 1424, 1387, 1326, 1136; MS (EI, 70 eV) m/z (%) 166 (M^+ , 0.45), 55 (100).

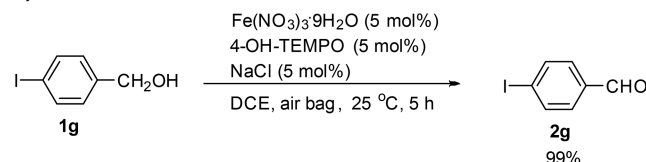
4-Nitrobenzaldehyde (2f) (jxg-5-147).



Following Typical Procedure I, the reaction of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.4 mg, 0.05 mmol), 4-OH-TEMPO (8.7 mg, 0.05 mmol), NaCl (2.9 mg, 0.05 mmol), and **1f** (154.7 mg, 99% purity, 1.0 mmol) in DCE (4.0 mL) for 5 h afforded **2f**²⁹ (151.4 mg, 91%) as a solid (eluent: petroleum ether/ethyl ether = 5/1 (400 mL)) (98% **2f** and 1.5% **3f**³⁰ were observed by NMR analysis of the

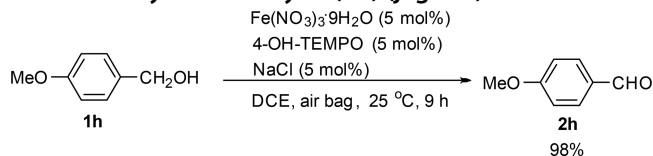
crude product using CH_2Br_2 as an internal standard). Mp 105.4–106.6 °C (ethyl acetate/petroleum ether; lit.²⁹ 104–106 °C); ^1H NMR (400 MHz, CDCl_3) δ 10.18 (s, 1H, CHO), 8.41 (d, J = 8.4 Hz, 2H, Ar–H), 8.10 (d, J = 8.0 Hz, 2H, Ar–H); ^{13}C NMR (100 MHz, CDCl_3) δ 190.3, 151.1, 140.0, 130.4, 124.2; IR (neat, cm^{-1}) 3106, 2850, 1703, 1604, 1532, 1343, 1324, 1286, 1194, 1103, 1006; MS (EI, 70 eV) m/z (%) 151 (M^+ , 100).

Typical Procedure II: 4-Iodobenzaldehyde (2g) (jxg-6-8).



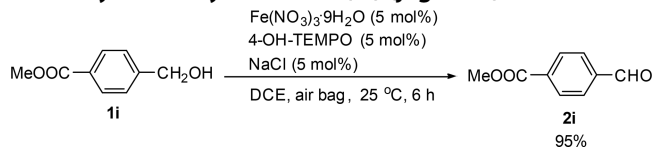
To a Schlenk tube were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.3 mg, 0.05 mmol), 4-OH-TEMPO (8.7 mg, 0.05 mmol), NaCl (2.9 mg, 0.05 mmol), **1g** (238.6 mg, 98% purity, 1.0 mmol), and DCE (4.0 mL) sequentially. The Schlenk tube was then connected with a gas bag (2 L (commercial size); could be expanded to 5 L) of air stirred at 25 °C until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 5/1) (5 h). The crude reaction mixture was filtered through a short column of silica gel (height, 2 cm; Φ , 3 cm), eluting with diethyl ether (3 $\times$ 25 mL). After evaporation, the residue was purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 20/1 (400 mL)) to afford **2g**³¹ (230.3 mg, 99%) as a solid. Mp 77.0–78.3 °C (ethyl acetate/petroleum ether; lit.³¹ 77–78 °C); ^1H NMR (400 MHz, CDCl_3) δ 9.96 (s, 1H, CHO), 7.92 (d, J = 8.4 Hz, 2H, Ar–H), 7.60 (d, J = 8.0 Hz, 2H, Ar–H); ^{13}C NMR (100 MHz, CDCl_3) δ 191.4, 138.4, 135.5, 130.8, 102.8; IR (neat, cm^{-1}) 2825, 2733, 1683, 1656, 1579, 1561, 1473, 1403, 1377, 1274, 1202, 1160, 1048, 1003; MS (EI, 70 eV) m/z (%) 232 (100).

4-Methoxybenzaldehyde (2h) (jxg-6-9).



Following Typical Procedure II, the reaction of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.1 mg, 0.05 mmol), 4-OH-TEMPO (8.7 mg, 0.05 mmol), NaCl (2.9 mg, 0.05 mmol), and **1h** (141.6 mg, 98% purity, 1.0 mmol) in DCE (4 mL) for 9 h afforded **2h**³² (133.6 mg, 98%) as an oil (eluent: petroleum ether/ethyl ether = 15/1 (650 mL)). ^1H NMR (400 MHz, CDCl_3) δ 9.88 (s, 1H, CHO), 7.84 (d, J = 8.8 Hz, 2H, Ar–H), 7.01 (d, J = 8.8 Hz, 2H, Ar–H), 3.89 (s, 3H, CH_3); ^{13}C NMR (100 MHz, CDCl_3) δ 190.8, 164.5, 131.9, 129.8, 114.2, 55.5; IR (neat, cm^{-1}) 2840, 2737, 1680, 1595, 1575, 1509, 1460, 1426, 1393, 1314, 1254, 1214, 1182, 1156, 1108, 1021; MS (EI, 70 eV) m/z (%) 136 (M^+ , 72.67), 135 (100).

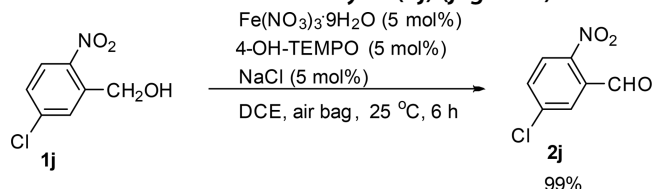
Methyl 4-Formylbenzoate (2i) (jxg-6-22).



Following Typical Procedure II, the reaction of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.4 mg, 0.05 mmol), 4-OH-TEMPO (8.7 mg, 0.05 mmol), NaCl (2.9 mg, 0.05 mmol), and **1i** (169.2 mg, 98% purity, 1.0 mmol) in DCE (4.0 mL) for 6 h afforded **2i**³³ (156.2

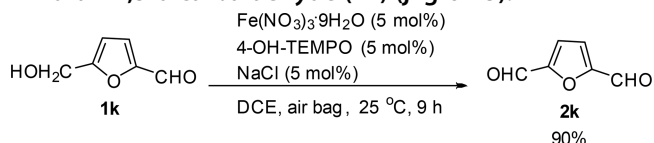
mg, 95%) as a solid (eluent: petroleum ether/ethyl acetate = 20/1). Mp 61.7–62.9 °C (ethyl acetate/petroleum ether; lit.³³ 61–63 °C); ¹H NMR (400 MHz, CDCl₃) δ 10.11 (s, 1H, CHO), 8.20 (d, *J* = 8.0 Hz, 2H, Ar–H), 7.96 (d, *J* = 8.4 Hz, 2H, Ar–H), 3.97 (s, 3H, CH₃); ¹³C NMR (100 MHz, CDCl₃) δ 191.6, 165.9, 139.0, 135.0, 130.1, 129.4, 52.5; IR (neat, cm^{−1}) 2962, 2887, 1722, 1682, 1575, 1502, 1434, 1391, 1280, 1199, 1106, 1012; MS (EI, 70 eV) *m/z* (%) 164 (M⁺, 59.96), 133 (100).

5-Chloro-2-nitrobenzaldehyde (2j) (jxg-6-23).



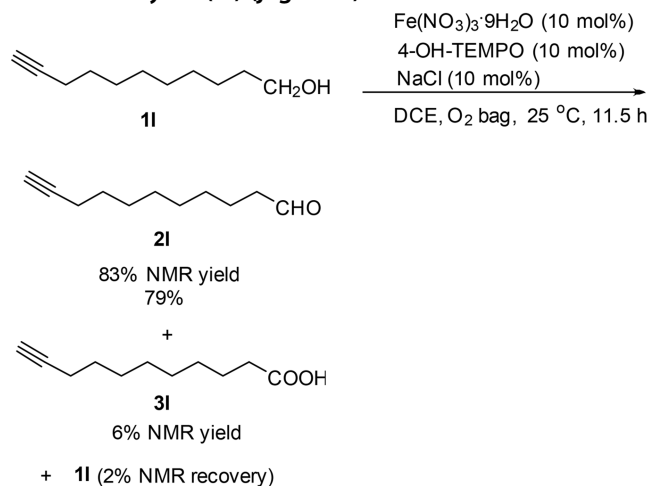
Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (20.4 mg, 0.05 mmol), 4-OH-TEMPO (8.6 mg, 0.05 mmol), NaCl (2.9 mg, 0.05 mmol), and **1j** (191.2 mg, 98% purity, 1.0 mmol) in DCE (4.0 mL) for 6 h afforded **2j**³⁴ (183.1 mg, 99%) as a solid (eluent: petroleum ether/ethyl acetate = 20/1). Mp 75.6–76.7 °C (ethyl acetate/petroleum ether; lit.³⁴ 74 °C (ethyl acetate/*n*-hexane)); ¹H NMR (400 MHz, CDCl₃) δ 10.42 (s, 1H, CHO), 8.13 (d, *J* = 8.8 Hz, 1H, Ar–H), 7.90 (d, *J* = 2.4 Hz, 1H, Ar–H), 7.73 (dd, *J*₁ = 8.4 Hz, *J*₂ = 2.4 Hz, 1H, Ar–H); ¹³C NMR (100 MHz, CDCl₃) δ 186.8, 147.4, 141.2, 133.4, 132.7, 129.6, 126.1; IR (neat, cm^{−1}) 3095, 1693, 1563, 1528, 1505, 1462, 1385, 1342, 1305, 1256, 1179, 1151, 1103, 1071; MS (EI, 70 eV) *m/z* (%) 187 ([M(³⁷Cl)]⁺, 0.32), 185 ([M(³⁵Cl)]⁺, 0.71), 155 (100).

Furan-2,5-dicarbaldehyde (2k) (jxg-6-25).



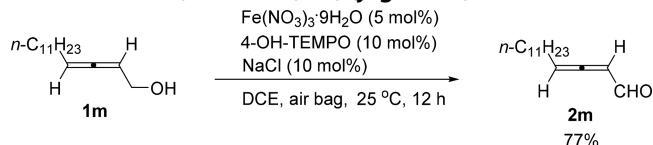
Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (20.3 mg, 0.05 mmol), 4-OH-TEMPO (8.7 mg, 0.05 mmol), NaCl (3.0 mg, 0.05 mmol), and **1k** (129.5 mg, 98% purity, 1.0 mmol) in DCE (4.0 mL) for 9 h afforded **2k**³⁵ (112.2 mg, 90%) as a solid (eluent: petroleum ether/ethyl acetate = 5/1 (800 mL)). Mp 110.0–111.4 °C (ethyl acetate/petroleum ether; lit.³⁵ 108–110 °C); ¹H NMR (400 MHz, CDCl₃) δ 9.87 (s, 2H, 2 × CHO), 7.37 (s, 2H, two protons of furanyl); ¹³C NMR (100 MHz, CDCl₃) δ 179.2, 154.1, 119.4; IR (neat, cm^{−1}) 3126, 3104, 1662, 1561, 1510, 1409, 1264, 1236, 1170, 1043, 1021, 1003; MS (EI, 70 eV) *m/z* (%) 124 (M⁺, 100).

10-Undecynal (2l) (jxg-6-21).



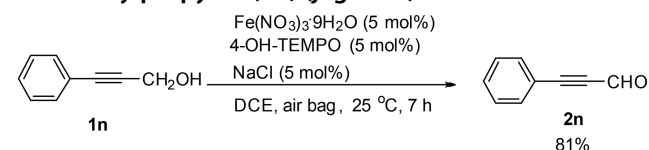
Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (40.5 mg, 0.1 mmol), 4-OH-TEMPO (17.4 mg, 0.1 mmol), NaCl (5.8 mg, 0.1 mmol), and **1l**³⁶ (168.3 mg, 1.0 mmol) in DCE (4.0 mL) for 11.5 h afforded **2l**³⁷ (131.8 mg, 79%) as an oil (eluent: petroleum ether/ethyl ether = 25/1 (500 mL)) (83% **2l**, 6% **3l**³⁸ and 2% **1l** were observed by NMR analysis of the crude product using CH₂Br₂ as an internal standard). ¹H NMR (400 MHz, CDCl₃) δ 9.77 (t, *J* = 1.8 Hz, 1H, CHO), 2.43 (td, *J*₁ = 7.3 Hz, *J*₂ = 1.6 Hz, 2H, CH₂), 2.18 (td, *J*₁ = 7.0 Hz, *J*₂ = 2.7 Hz, 2H, CH₂), 1.95 (t, *J* = 2.4 Hz, 1H, ≡CH), 1.63 (quint, *J* = 7.2 Hz, 2H, CH₂), 1.52 (quint, *J* = 7.3 Hz, 2H, CH₂), 1.45–1.25 (m, 8H, 4 × CH₂); ¹³C NMR (100 MHz, CDCl₃) δ 202.8, 84.6, 68.1, 43.8, 29.1, 29.0, 28.8, 28.5, 28.3, 21.9, 18.3; IR (neat, cm^{−1}) 3291, 2929, 2856, 2720, 2117, 1723, 1462, 1410, 1391, 1353; MS (EI, 70 eV) *m/z* (%) 166 (M⁺, 0.15), 81 (100).

Pentadeca-2,3-dienal (2m) (jxg-6-26).



Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (20.4 mg, 0.05 mmol), 4-OH-TEMPO (17.2 mg, 0.1 mmol), NaCl (5.9 mg, 0.1 mmol), and **1m**³⁹ (225.1 mg, 1.0 mmol) in DCE (4.0 mL) for 12 h afforded **2m** (172.8 mg, 77%) as an oil (eluent: petroleum ether/ethyl ether = 50/1). ¹H NMR (400 MHz, CDCl₃) δ 9.49 (d, *J* = 7.6 Hz, 1H, CHO), 5.85–5.73 (m, 2H, HC=C=CH), 2.19 (qd, *J*₁ = 7.1 Hz, *J*₂ = 3.0 Hz, 2H, CH₂), 1.48 (quint, *J* = 7.2 Hz, 2H, CH₂), 1.40–1.20 (m, 16H, 8 × CH₂), 0.88 (t, *J* = 6.8 Hz, 3H, CH₃); ¹³C NMR (100 MHz, CDCl₃) δ 219.1, 192.3, 98.6, 96.3, 31.9, 29.57, 29.55, 29.5, 29.29, 29.25, 28.9, 28.8, 27.4, 22.6, 14.1; IR (neat, cm^{−1}) 2922, 2853, 1943, 1689, 1463, 1357, 1108, 1084; MS (EI, 70 eV) *m/z* (%) 222 (M⁺, 3.75), 81 (100); HRMS calcd for C₁₅H₂₆O (M⁺) 222.1984, found 222.1982.

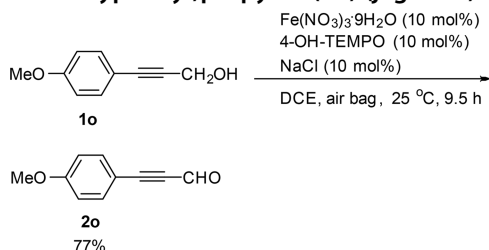
3-Phenylpropynal (2n) (jxg-6-16).



Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (20.1 mg, 0.05 mmol), 4-OH-TEMPO (8.6 mg, 0.05

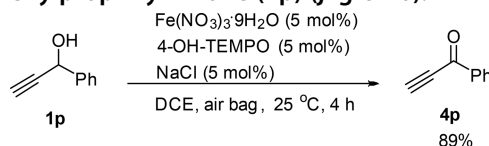
mmol), NaCl (2.9 mg, 0.05 mmol), and **1n** (132.3 mg, 1.0 mmol) in DCE (4.0 mL) for 7 h afforded **2n**^{12a} (106.0 mg, 81%) as an oil (eluent: petroleum ether/ethyl ether = 20/1 (400 mL)). ¹H NMR (400 MHz, CDCl₃) δ 9.42 (s, 1H, CHO), 7.63–7.57 (m, 2H, Ar–H), 7.52–7.45 (m, 1H, Ar–H), 7.44–7.37 (m, 2H, Ar–H); ¹³C NMR (100 MHz, CDCl₃) δ 176.8, 133.2, 131.2, 128.7, 119.3, 95.1, 88.3; IR (neat, cm^{−1}) 2853, 2239, 2185, 1653, 1488, 1443, 1387, 1259, 1160, 1069, 1027, 1002; MS (EI, 70 eV) *m/z* (%) 130 (100).

3-(4-Methoxyphenyl)propynal (**2o**) (jxg-6-19).



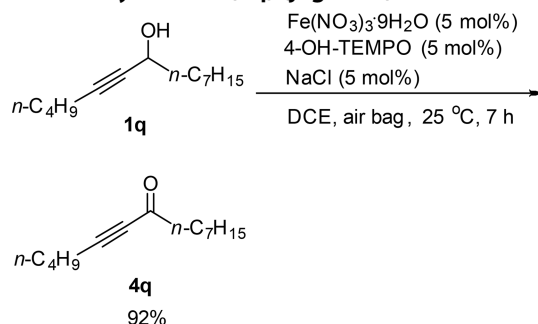
Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (40.4 mg, 0.1 mmol), 4-OH-TEMPO (17.2 mg, 0.1 mmol), NaCl (5.9 mg, 0.1 mmol), and **1o**⁴⁰ (162.1 mg, 1.0 mmol) in DCE (4.0 mL) for 9.5 h afforded **2o**⁴¹ (123.6 mg, 77%) as a solid (eluent: petroleum ether/ethyl ether = 20/1 (500 mL)). Mp 47.1–47.9 °C (ethyl acetate/petroleum ether; lit.⁴¹ 47–48 °C); ¹H NMR (400 MHz, CDCl₃) δ 9.39 (s, 1H, CHO), 7.56 (d, *J* = 8.8 Hz, 2H, Ar–H), 6.91 (d, *J* = 8.8 Hz, 2H, Ar–H), 3.85 (s, 3H, CH₃); ¹³C NMR (100 MHz, CDCl₃) δ 176.7, 162.0, 135.4, 114.4, 111.0, 96.5, 88.7, 55.4; IR (neat, cm^{−1}) 2933, 2839, 2249, 2177, 1641, 1596, 1566, 1506, 1463, 1416, 1387, 1302, 1253, 1174, 1112, 1020; MS (EI, 70 eV) *m/z* (%) 160 (M⁺, 100).

1-Phenylprop-2-yn-1-one (**4p**) (jxg-6-10).



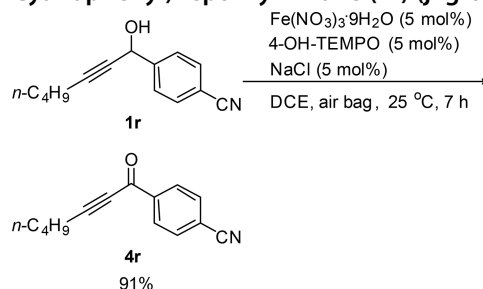
Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (20.4 mg, 0.05 mmol), 4-OH-TEMPO (8.5 mg, 0.05 mmol), NaCl (2.9 mg, 0.05 mmol), and **1p** (135.5 mg, 0.5 mmol) in DCE (4.0 mL) for 4 h afforded **4p**^{12b} (115.5 mg, 89%) as a solid (eluent: petroleum ether/ethyl ether = 20/1 (400 mL)). Mp 50.1–51.1 °C (ethyl acetate/petroleum ether; lit.^{12b} 50–51 °C); ¹H NMR (400 MHz, CDCl₃) δ 8.20–8.13 (m, 2H, Ar–H), 7.64 (t, *J* = 7.4 Hz, 1H, Ar–H), 7.50 (t, *J* = 7.6 Hz, 2H, Ar–H), 3.47 (s, 1H, ≡CH); ¹³C NMR (100 MHz, CDCl₃) δ 177.4, 136.0, 134.5, 129.6, 128.6, 80.8, 80.1; IR (neat, cm^{−1}) 3230, 2090, 1639, 1594, 1576, 1451, 1415, 1313, 1259, 1172, 1030, 1003; MS (EI, 70 eV) *m/z* (%) 130 (M⁺, 53.22), 102 (100).

Tetradec-5-yn-7-one (**4q**) (jxg-6-34).



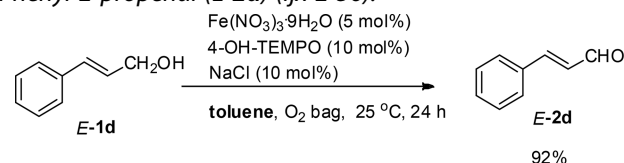
Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (20.2 mg, 0.05 mmol), 4-OH-TEMPO (8.6 mg, 0.05 mmol), NaCl (2.9 mg, 0.05 mmol), and **1q**⁴² (210.6 mg, 1.0 mmol) in DCE (4.0 mL) for 7 h afforded **4q**⁴³ (192.4 mg, 92%) as an oil (eluent: petroleum ether/ethyl ether = 40/1 (500 mL)). ¹H NMR (400 MHz, CDCl₃) δ 2.52 (t, *J* = 7.6 Hz, 2H, CH₂), 2.37 (t, *J* = 7.0 Hz, 2H, CH₂), 1.66 (quint, *J* = 7.1 Hz, 2H), 1.62–1.53 (m, 2H, CH₂), 1.49–1.38 (m, 2H, CH₂), 1.37–1.20 (m, 8H, 4 × CH₂), 0.93 (t, *J* = 7.4 Hz, 3H, CH₃), 0.88 (t, *J* = 6.8 Hz, 3H, CH₃); ¹³C NMR (100 MHz, CDCl₃) δ 188.5, 94.1, 80.8, 45.5, 31.6, 29.7, 28.93, 28.87, 24.1, 22.5, 21.9, 18.6, 14.0, 13.4; IR (neat, cm^{−1}) 2957, 2928, 2858, 2212, 1672, 1463, 1407, 1378, 1356, 1325, 1299, 1247, 1226, 1162, 1106, 1053, 1000; MS (EI, 70 eV) *m/z* (%) 208 (M⁺, 0.32), 109 (100).

1-(4-Cyanophenyl)hept-2-yn-1-one (**4r**) (jxg-6-35).

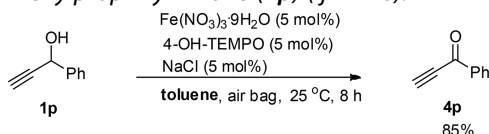


Following Typical Procedure II, the reaction of Fe(NO₃)₃·9H₂O (20.3 mg, 0.05 mmol), 4-OH-TEMPO (8.6 mg, 0.05 mmol), NaCl (3.0 mg, 0.05 mmol), and **1r**⁴⁴ (213.0 mg, 1.0 mmol) in DCE (4.0 mL) for 7 h afforded **4r**⁴⁵ (192.5 mg, 91%) as an oil (eluent: petroleum ether/ethyl ether = 20/1 (500 mL) to 15/1 (300 mL)). ¹H NMR (400 MHz, CDCl₃) δ 8.23 (d, *J* = 8.8 Hz, 2H, Ar–H), 7.80 (d, *J* = 8.4 Hz, 2H, Ar–H), 2.55 (t, *J* = 7.0 Hz, 2H, CH₂), 1.73–1.64 (m, 2H, CH₂), 1.57–1.45 (m, 2H, CH₂), 0.98 (t, *J* = 7.2 Hz, 3H, CH₃); ¹³C NMR (100 MHz, CDCl₃) δ 176.3, 139.5, 132.3, 129.7, 117.8, 116.8, 99.0, 79.2, 29.6, 22.0, 18.9, 13.4; IR (neat, cm^{−1}) 2959, 2933, 2871, 2231, 2198, 1647, 1605, 1566, 1463, 1404, 1309, 1293, 1257, 1175, 1110, 1017; MS (EI, 70 eV) *m/z* (%) 211 (M⁺, 5.56), 169 (100).

Aerobic Oxidation of Alcohols in Toluene. 1. (*E*)-3-Phenyl-2-propenal (**E-2d**) (ljx-2-56).

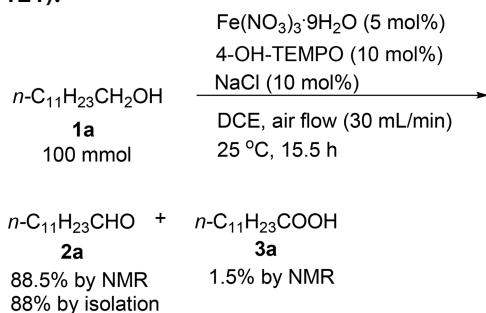


To a Schlenk tube were added Fe(NO₃)₃·9H₂O (20.4 mg, 0.05 mmol), 4-OH-TEMPO (17.2 mg, 0.1 mmol), NaCl (5.7 mg, 0.1 mmol), **E-1d** (134.5 mg, 1.0 mmol), and toluene (4.0 mL) sequentially under an atmosphere of oxygen from a 2 L (commercial size) gas bag. The Schlenk tube was then stirred at 25 °C until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 3/1) (24 h). The crude reaction mixture was filtered through a short column of silica gel (height, 2 cm; Φ, 3 cm), eluting with diethyl ether (3 × 25 mL). After evaporation, the residue was purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 10/1) to afford **E-2d**¹⁹ (122.5 mg, 92%) as an oil (95% **E-2d** was observed by NMR analysis of the crude product using CH₂Br₂ as an internal standard).

2. 1-Phenylprop-2-yn-1-one (**4p**) (ljx-2-46).

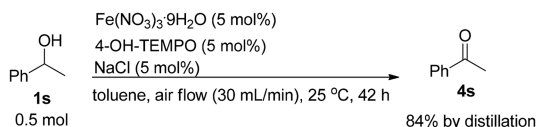
To a Schlenk tube were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.2 mg, 0.05 mmol), 4-OH-TEMPO (8.8 mg, 0.05 mmol), NaCl (2.7 mg, 0.05 mmol), **1p** (132.1 mg, 1.0 mmol), and toluene (4.0 mL) sequentially. The Schlenk tube was then connected with a 2 L (commercial size) gas bag of air stirred at 25 °C until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 5/1) (8 h). The crude reaction mixture was filtered through a short column of silica gel (height, 2 cm; Φ , 3 cm), eluting with diethyl ether (3 $\times$ 25 mL). After evaporation, the residue was purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 20/1 (200 mL)) to afford **4p**³¹ (110.5 mg, 85%) as a solid.

Reaction on a 100 mmol Scale with a Slow Flow of Air (jxg-6-121).



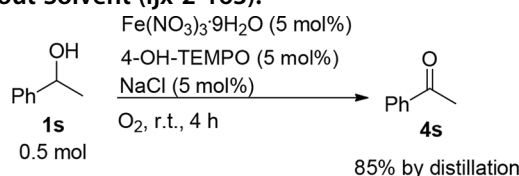
To a 1 L three-neck flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (2.0210 g, 5.0 mmol), 4-OH-TEMPO (1.7212 g, 10.0 mmol), NaCl (0.5846 g, 10.0 mmol), and DCE (200 mL) sequentially. After 10 min of stirring, **1a** (18.8203 g, 99% purity, 100.0 mmol) and DCE (100 mL) were added. A slow flow of air directly from an air cylinder (99.99%, 30 mL/min) was then connected to the flask (Figure 1). The resulting mixture was stirred at 25 °C until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 5/1) (15.5 h). The crude reaction mixture was filtered through a short column of silica gel (height, 3 cm; Φ , 7.5 cm), eluting with ethyl ether (300 mL). After evaporation, the residue was purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 100/1 (3.2 L)) to afford **2a** (16.3054 g, 88%) as an oil (88.5% **2a** and 1.5% **3a** were observed by NMR analysis of the crude product using CH_2Br_2 as an internal standard). ¹H NMR (400 MHz, CDCl_3) δ 9.76 (t, J = 1.6 Hz, 1H, CHO), 2.42 (td, J_1 = 7.3 Hz, J_2 = 1.7 Hz, 2H, CH_2), 1.63 (quint, J = 7.3 Hz, 2H, CH_2), 1.38–1.16 (m, 16H, 8 $\times$ CH_2), 0.88 (t, J = 7.0 Hz, 3H, CH_3); ¹³C NMR (100 MHz, CDCl_3) δ 202.9, 43.9, 31.9, 29.6, 29.4, 29.32, 29.29, 29.1, 22.6, 22.1, 14.1.

Reaction on a 0.5 mol Scale with a Slow Flow of Air (ljx-2-115).



To a 500 mL three-neck flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (10.1407 g, 25.1 mmol), 4-OH-TEMPO (4.3114 g, 25.0 mmol), NaCl (1.4754 g, 25.2 mmol), **1s** (61.0180 g, 0.5 mol), and toluene (100 mL) sequentially. A slow flow of air directly from

an air cylinder (30 mL/min) was then connected to the flask, and the temperature increased to 35 °C. The resulting mixture was stirred at 25 °C until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 5/1) (42 h). The liquid layer was transferred to a 500 mL flask for distillation, and the residue was washed with DCM (10 mL $\times$ 3) and added to the distillation flask, after which the resulting mixture was distilled at reduced pressure to afford **4s**^{12a} in 84% yield (50.2509 g) as a yellow oil. ¹H NMR (300 MHz, CDCl_3) δ 7.96 (d, J = 7.2 Hz, 2H, Ar–H), 7.57 (t, J = 7.2 Hz, 1H, Ar–H), 7.47 (t, J = 7.2 Hz, 2H, Ar–H), 2.61 (s, 3H, CH_3); ¹³C NMR (75 MHz, CDCl_3) δ 198.1, 137.1, 133.1, 128.5, 128.3, 26.6; IR (neat, cm^{-1}) 3062, 1682, 1599, 1582, 1449, 1429, 1359, 1266, 1180, 1079, 1025; MS (EI, 70 eV) m/z (%) 120 (M^+ , 34.07), 105 (100).

Reaction of **1s** on a 0.5 mol Scale with Pure Oxygen without Solvent (ljx-2-163).

To a 500 mL three-neck flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (10.1058 g, 25.0 mmol), 4-OH-TEMPO (4.2969 g, 24.9 mmol), NaCl (1.5068 g, 25.8 mmol), and **1s** (61.1377 g, 0.5 mol) sequentially, the atmosphere was replaced by pure oxygen twice from a gas bag (commercial size: 2 L). The resulting mixture was stirred at room temperature under the atmosphere of pure oxygen until completion of the reaction as monitored by NMR analysis (**Caution:** the reaction temperature would rise up to 62 °C and an ice water bath was used until the temperature dropped down to 25 °C, repeat this manipulation when necessary). The resulting mixture was transferred to a 250 mL flask for distillation in reduced pressure to afford **4s** in 85% yield (51.1496 g, yellow oil, bp 70–72 °C/7.0 Torr (lit. 73 °C/7.0 Torr⁴⁶)).

■ ASSOCIATED CONTENT

S Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.oprd.8b00374.

¹H and ¹³C NMR spectra for all of the products (PDF)

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Notes

The authors declare no competing financial interest.

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